Task 1 – Review of the Existing Architecture

1.Identified Components of the Workload

Based on the provided background (two-tier web application migrating from on-premises to AWS), the workload consists of the following components:

Frontend Web Application Server

- Hosts the user-facing application
- Handles HTTP/HTTPS requests
- Contains presentation logic
- May run on a virtual machine or physical server

Backend Database Server

- Stores application data
- Handles queries from the frontend
- Likely self-managed (e.g., MySQL or PostgreSQL)

Networking Infrastructure

- Internal network configuration connecting frontend to backend
- Firewall rules controlling inbound/outbound traffic

On-Premises Hosting Environment

- Physical or virtual servers
- Local storage systems
- Power and cooling dependencies

2.Potential Risks and Weaknesses

Single Point of Failure

If the application and database are hosted in a single location or Availability Zone, any hardware or network failure could result in complete system downtime.

No Backup Strategy

There is no mention of automated database backups.
This increases the risk of permanent data loss in case of failure or corruption.

Lack of High Availability

No redundancy mechanisms (e.g., multi-zone deployment or replication) are indicated.

Limited Scalability

On-prem infrastructure does not dynamically scale with traffic demand, leading to:

- Performance degradation during peak usage
- Underutilized resources during low traffic

Potential Security Misconfigurations

- Open firewall/security group rules (e.g., 0.0.0.0/0 access)
- No centralized identity and access management
- Possible lack of encryption for data at rest and in transit

Absence of Monitoring and Logging

Without centralized monitoring, system issues may go undetected, increasing recovery time.

Manual Operational Processes

Manual deployments and updates increase the likelihood of human error.

Task 2 – AWS Well-Architected Framework Assessment

Pillar	Observation	Improvement Recommendation	Supporting AWS Service
Operational Excellence	Deployment and system management appear manual, with no centralized monitoring or automation.	Implement Infrastructure as Code and centralized monitoring to improve operational consistency and visibility.	AWS CloudFormation, Amazon CloudWatch
Security	Potential exposure due to open firewall/security rules and lack of encryption strategy.	Enforce least privilege access, encrypt data at rest and in transit, and restrict inbound traffic.	AWS IAM, AWS KMS, AWS WAF

Reliability	Single-location deployment introduces a single point of failure and no automated backup strategy.	Deploy across multiple Availability Zones and enable automated backups and replication.	Amazon RDS Multi-AZ, AWS Backup
Performance Efficiency	Static resource allocation limits responsiveness to varying traffic loads.	Implement Auto Scaling and use managed database services to optimize performance dynamically.	EC2 Auto Scaling, Amazon RDS
Cost Optimization	On-premises infrastructure may be overprovisioned and lacks real-time cost visibility.	Use right-sizing tools and cost monitoring to optimize cloud spending.	AWS Cost Explorer, AWS Compute Optimizer

Task 3 – AWS Cloud Adoption Framework (CAF) Assessment

Business Perspective

The organization demonstrates moderate readiness from a business standpoint, as evidenced by its decision to migrate a two-tier application from on-premises infrastructure to AWS. This indicates executive recognition of modernization and scalability needs. However, there is no evidence of a formal cloud business case, defined return on investment (ROI), or measurable Key Performance Indicators (KPIs) tied to migration outcomes. The absence of structured financial governance introduces risks such as cost overruns and unclear value realization. To improve readiness, the organization should develop a detailed cloud business strategy outlining expected ROI, operational efficiency improvements, and cost optimization objectives. Establishing clear KPIs and implementing budgeting controls and cost allocation models will strengthen financial accountability. These actions will ensure cloud adoption aligns with strategic objectives and delivers measurable business value.

People Perspective

The organization appears to have limited demonstrated cloud expertise, suggesting low to moderate readiness from a people perspective. While the migration initiative indicates

willingness to adopt new technologies, there is no indication of structured cloud training programs, defined cloud roles, or a Cloud Center of Excellence (CCoE). This skills gap may hinder effective implementation and long-term management of cloud services. To enhance readiness, the organization should invest in structured training and certification programs for technical teams. Establishing a CCoE would promote governance consistency and knowledge sharing. Clearly defining roles for cloud architects, DevOps engineers, and security personnel will improve accountability. Encouraging collaboration across departments will support a DevOps culture and sustainable cloud operations.

Governance Perspective

Governance readiness appears limited due to the absence of clearly defined cloud policies, access controls, and cost management frameworks. The migration plan does not reference standardized approval processes, tagging strategies, or compliance monitoring mechanisms. Without structured governance, the organization risks resource sprawl, budget overruns, and inconsistent security enforcement. To address these gaps, centralized account management and defined policy controls should be implemented. A standardized tagging strategy for cost allocation and auditing is recommended. Financial monitoring mechanisms and formal change management procedures should also be established. Strengthening governance will ensure transparency, regulatory compliance, and alignment with enterprise risk management practices.

Platform Perspective

From a platform perspective, the organization demonstrates moderate readiness due to its willingness to migrate infrastructure to the cloud. However, the current two-tier design suggests limited adoption of cloud-native principles such as automation, scalability, and fault tolerance. There is no indication of Infrastructure as Code (IaC) or automated provisioning. To improve readiness, the organization should adopt IaC tools to enable repeatable deployments and reduce configuration errors. Designing a segmented network architecture and leveraging managed services will reduce operational burden.
Implementing auto scaling and load balancing will improve availability and performance.
These platform improvements will provide a resilient and scalable foundation.

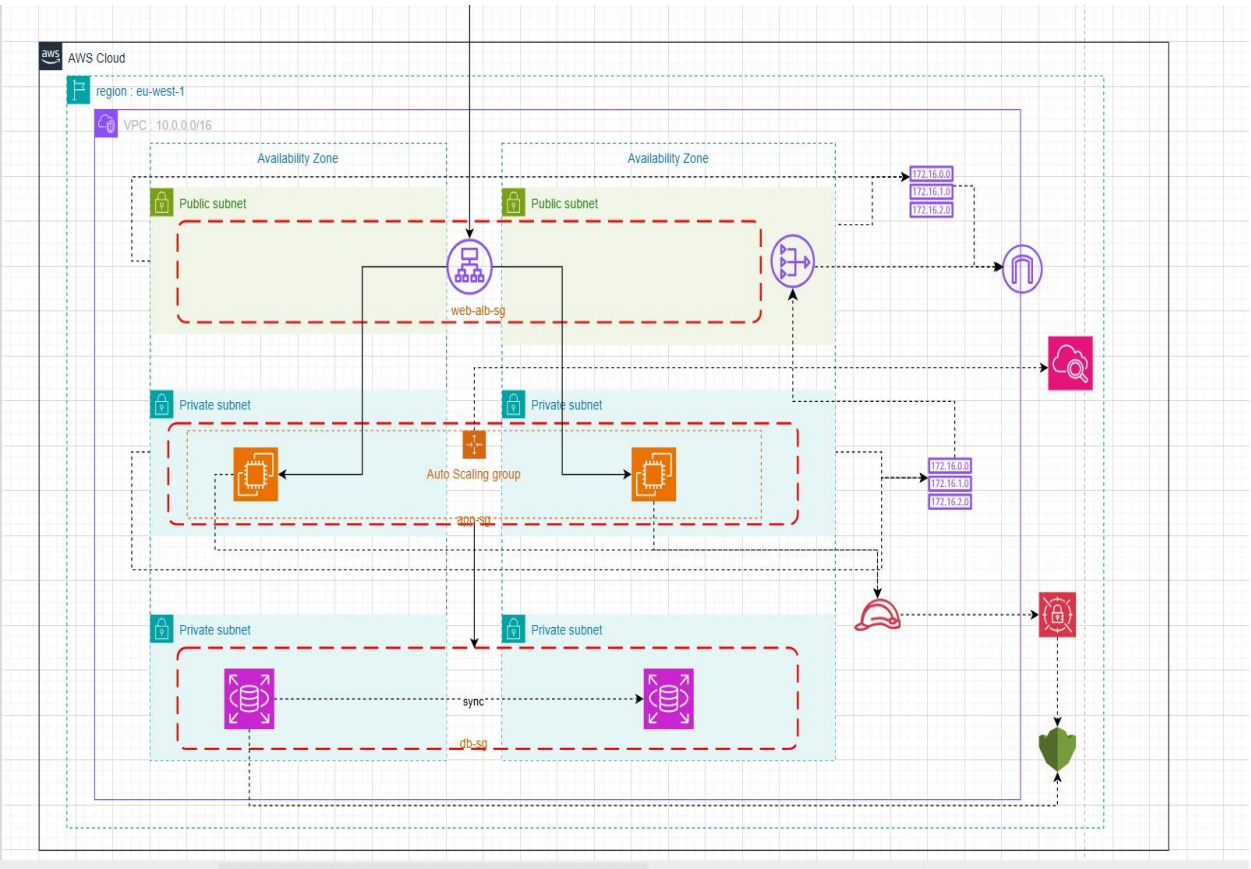
Security Perspective

Security readiness appears relatively low due to limited evidence of structured identity management, encryption standards, and centralized monitoring. The migration scenario does not specify encryption practices or least-privilege access policies, increasing exposure to data breaches and compliance risks. To strengthen security posture, the organization should enforce role-based access controls and multi-factor authentication for privileged accounts. Data encryption at rest and in transit must be implemented. Logging and auditing mechanisms should be established to monitor system activity. Embedding security controls into the migration process will reduce risk exposure and support regulatory compliance.

Operations Perspective

Operational readiness is currently limited due to the absence of clearly defined monitoring, incident response, and disaster recovery procedures. The existing architecture suggests reliance on manual processes, increasing the likelihood of downtime and human error. There is no evidence of automated backup strategies or performance monitoring frameworks. To enhance operational maturity, centralized monitoring and logging solutions should be implemented. Defined Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO) must be established. Incident response procedures should be documented and tested regularly. Strengthening operational practices will ensure the cloud environment remains resilient, observable, and continuously optimized.

Task 4 – Improved AWS Architecture Design



The diagram illustrates a refactored two-tier web application architecture deployed within the AWS Cloud. The infrastructure is hosted in the eu-west-1 region inside a custom Virtual Private Cloud (VPC) with CIDR block 10.0.0.0/16. The VPC spans two Availability Zones to ensure high availability and fault tolerance. At the top layer, public subnets in each Availability Zone host an Application Load Balancer (ALB). The ALB serves as the internet-facing entry point and distributes incoming user traffic across application servers deployed in private subnets. The application tier consists of EC2 instances configured within an Auto Scaling Group. These instances are distributed across both Availability Zones to eliminate single points of failure and automatically scale based on demand. Because the EC2 instances are deployed in private subnets, they are not directly accessible from the internet, improving the security posture. The database tier is implemented using Amazon RDS in a Multi-AZ configuration. A primary database instance runs in one Availability Zone while a standby replica is synchronized in the second Availability Zone. In the event of failure, automatic failover ensures minimal downtime and high data durability. Additional supporting services such as monitoring, logging, and security controls are integrated to enhance operational visibility and governance. This architecture aligns with the AWS Well-Architected Framework by addressing Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization from inception.

Reflection – AWS Well-Architected & Cloud Adoption Lab

This lab strengthened my understanding of how structured frameworks guide effective cloud architecture design. By applying the AWS Well-Architected Framework, I learned how to critically evaluate a workload across Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization pillars. Rather than focusing only on infrastructure components, I gained insight into how architectural decisions impact long-term scalability, resilience, and risk management. The Cloud Adoption Framework further expanded my perspective by highlighting the importance of organizational readiness, governance structures, skills development, and strategic alignment during cloud transformation. I realized that successful migration is not solely a technical exercise but a coordinated effort involving business strategy, people capability, and operational maturity. Designing the improved architecture reinforced the importance of eliminating single points of failure and embedding security, automation, and monitoring from the outset. Overall, this lab enhanced my ability to think systematically and architect solutions aligned with AWS best practices.